

THE UNIVERSITY OF HONG KONG
HKU SPACE Community College

CCIT4033 - Introduction to Database Systems

Individual Assignment

Due Date : 29/Mar/2023 18:00
Total Marks : 100
Weight : 15% of the total assessment of this course
Submission : Softcopy (through SOUL) – MS Word format ONLY

Requirements:

This is an individual assignment. You may discuss with each other, but you must write your own answers.

Hand-drawn diagrams are acceptable. Make sure they are clearly readable.

Part 1

Answer Q1-Q2 with reference to the following Employee Register for the Talent Union Ltd.

Talent Union Ltd
Employee Register

Name YEUNG Hong Department Sales Staff No. 0862

No.	Dependent name	Relationship	Date of birth
1	CHING Wu	Spouse	17/05/1981
2	YEUNG Tai Long	Son	26/12/2003
3	YEUNG Siu Miu	Daughter	13/02/2006

Number of dependents entitled for medical benefits: 2

Medical benefits entitlement

1. Spouse of the employee at any age
2. Child of the employee of age below 18

** sample data for illustration only*



Assumptions:

- Spouse of the employee at any age entitles medical benefits.
- Child of the employee of age below 18 entitles medical benefits.
- There is no limit for the number of dependents of an employee.
- There are no differences in medical benefits entitlement for staff members of different job positions.

Q1. Draw a Crow's Foot ER diagram to represent the data model for the Employee Register, including all attributes, relationships, primary keys and foreign keys that can be identified on the record. Add new attributes and include any assumptions where necessary. (20 marks)

Q2. Referring to the ERD that you have developed in Q1, there may have attribute(s) likely to be redundant in the entities. List the redundant attribute(s). (2 marks)
Describe one advantage and one disadvantage of having redundant attribute(s) in the database design. (8 marks)

Part 2

Answer Q3-Q5 with reference to the BIKE_RENTAL_RECORD table below.

BIKE_RENTAL_RECORD

<u>Rental_date</u>	Cus_name	<u>Cus_phone</u>	<u>Bike_no</u>	Description	Bike_type	Charge	Status
02 Mar	Charles LOPEZ	9003 0369	S047	Bright red racer	Speedy	\$220	Returned
02 Mar	Charles LOPEZ	9003 0369	K11	Kid of the road	Kid	\$160	Returned
02 Mar	Daniel Wong	6517 0394	S008	Swift racer	Speedy	\$220	Lost
09 Mar	YAN Ho Ming	5439 4466	L01532	Legend 1532	Learner	\$190	Active
09 Mar	YAN Ho Ming	5439 4466	K11	Kid of the road	Kid	\$160	Active

where,

- (Rental_date, Cus_phone, Bike_no) is the Primary Key
- Cus_phone determines Cus_name
- Bike_no determines Description, Bike_type, Charge
- Bike_type determines Charge
- (Rental_date, Cus_phone, Bike_no) determines Status

Q3. Using the BIKE_RENTAL_RECORD table above, identify one example each for insertion, deletion and modification anomalies. (9 marks)

Q4. Draw the dependency diagram for the BIKE_RENTAL_RECORD table above. Identify all dependencies including functional, partial and transitive dependencies. (6 marks)

Q5. Normalize the BIKE_RENTAL_RECORD table to 3NF, showing all intermediate steps. (20 marks)

Part 3

Answer Q6-Q10 with reference to the SPECIALIST, TREATMENT, and BOOKING tables with which a health care centre assigns specialists to treatments for clients.

SPECIALIST

Primary Key: SPECIALIST_ID

 SPECIALIST_ID	SPECIALIST_NAME	SPECIALIST_TYPE
DY12	Dave Yu	Cognitive therapist
EH83	Edward Hung	Cognitive therapist
PC52	Paul Chung	Addiction therapist
SL92	Sara Lau	Addiction therapist
TM38	Tommy Ma	Cognitive therapist
WT60	Winnie Tong	Behavioural therapist

TREATMENT

Primary Key: TREATMENT_ID

 TREATMENT_ID	TREATMENT_DESC	TREATMENT_TYPE
AT330	Get rid of gambling	Addiction therapist
BT38	Be a nice father	Behavioural therapist
CT26	Cognitive for elderly	Cognitive therapist
CT854	Cognitive training	Cognitive therapist

BOOKING

Primary Key: BOOK_ID

Foreign Keys: BOOK_SPECIALIST, BOOK_TREATMENT

 BOOK_ID	BOOK_TREATMENT	BOOK_SPECIALIST	BOOK_DATE	BOOK_TIME	BOOK_HRS
B367	AT330	PC52	2023/3/6	10:30	1.5
B368	CT854	WT60	2023/3/6	11:00	2
B369	CT26	EH83	2023/3/6	11:30	1.5
B370	AT330	SL92	2023/3/6	11:30	1.5
B371	BT38	PC52	2023/3/6	14:00	1
B372	BT38	WT60	2023/3/6	14:30	2
B373	CT26	EH83	2023/3/6	15:00	2

Q6. Does the BOOKING table exhibit referential integrity? Explain your reasons (5 marks)

Q7. What is the result of SPECIALIST LEFT OUTER JOIN BOOKING? (8 marks)

Q8. Write an SQL statement to answer the question: *'Compute the total treatment hours by specialists in the sequence of specialist name.'* The resulting table is as follows, sorted by specialist name. (8 marks)

SPECIALIST NAME	TOTAL HOURS
Edward Hung	3.5
Paul Chung	2.5
Sara Lau	1.5
Winnie Tong	4

Q9. Write an SQL statement to answer the question: *'Are there any bookings not handled by appropriate specialists, i.e. specialist type not matching treatment type?'* The resulting table is as follows. (8 marks)

BOOK_ID	TREATMENT_DESC	TREATMENT_TYPE	SPECIALIST_ID	SPECIALIST_TYPE	BOOK_DATE	BOOK_TIME	BOOK_HR
B368	Cognitive training	Cognitive therapist	Winnie Tong	Behavioural therapist	2023/3/6	11:00	2
B371	Be a nice father	Behavioural therapist	Paul Chung	Addiction therapist	2023/3/6	14:00	1

Q10. In view of the possible problem as per Q9, suggest a way to prevent such booking. You may consider running query to highlight this or changing the data model to prevent so. Explain your suggestion. (6 marks)

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